

Future Interns – Cyber Security Task 2 (2026)

Phishing Email Detection & Awareness Report

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Sample: Sample-01 / sample-10.eml

Classification: PHISHING

Confidence: HIGH

Overall Risk: HIGH

1. Executive Summary

This report documents a static analysis of a publicly sourced phishing email sample collected by the rf-peixoto/phishing_pot honeypot project. The message impersonates the Microsoft account team and claims that an unusual sign-in occurred from Russia/Moscow.

Multiple independent indicators support a high-confidence phishing classification: Microsoft brand impersonation; a non-Microsoft From domain; a different Reply-To address; a different Return-Path domain; SPF none; DKIM none; DMARC permerror; urgency and fear-based language; a Gmail destination for the Report The User action; and an unrelated external HTTP tracking resource.

2. Objective

Analyze a real phishing-email sample, examine headers, identify phishing indicators and IOCs, inspect links without visiting them, classify risk, explain the attack simply, and produce practical awareness guidance.

3. Scope and Safety

Analysis was limited to the local .eml file. No embedded URL was visited, no reply was sent, no account was accessed, and no external infrastructure was contacted. The external tracking URL is defanged in the report.

4. Sample Information

Source: rf-peixoto/phishing_pot. Sample: email/sample-10.eml. Local file: samples/sample-01/sample-10.eml. SHA-256: 4fbf4c3d80aba156c59004c12c83ff53dd64c9cf7b7a6029e98fe1da0760783a. Message date: Fri, 8 Sep 2023 05:47:04 +0000. Recipient in sample: phishing@pot.

5. Analysis Methodology

Workflow: Collect → Preserve → Inspect → Extract → Correlate → Classify → Recommend.

Tools: Kali GNU/Linux, grep, sed, sort, head, sha256sum, cat, and locally stored evidence artifacts.

6. Header Analysis

From: Microsoft account team ,_<no-reply@access-accsecurity.com>. Reply-To: sotrecognizd@gmail.com. Return-Path: bounce@thcultarfdes.co.uk. Subject: Microsoft account unusual signin activity. X-Sender-IP: 89.144.44.2. Message-ID was observed with an empty value.

The message presents itself as Microsoft, but the technical identities do not align: the display identity claims Microsoft, the From domain is access-accsecurity.com, the SMTP envelope domain is thcultarfdes.co.uk, and the Reply-To is a Gmail address.

7. Authentication Analysis

Authentication-Results: spf=none; smtp.mailfrom=thcultarfdes.co.uk; dkim=none; header.d=none; dmarc=permerror; header.from=access-accsecurity.com.

Received-SPF states that thcultarfdes.co.uk does not designate permitted sender hosts. These results do not independently prove maliciousness, but combined with the identity mismatches they provide strong supporting evidence.

8. Link and Destination Analysis

Four unique destinations were extracted without opening them. External resource: [https://thebandalisty\[.\]com/track/\[REDACTED\]](https://thebandalisty[.]com/track/[REDACTED]). Three mailto destinations use sotrecognizd@gmail.com, including the Report The User action. The mismatch between the apparent Microsoft security action and the Gmail destination is a critical indicator.

9. Social Engineering Analysis

Techniques include brand impersonation, fear and urgency, specific-looking technical details, action pressure, and a security-notification disguise. The email claims a new login from Russia/Moscow and provides IP 103.225.77.255. This is the alleged login IP in the message, not the sender IP identified by the headers.

10. Risk Classification

Classification: PHISHING. Confidence: HIGH. Overall Risk: HIGH.

The classification is based on multiple independent indicators: Microsoft impersonation, non-Microsoft From domain, Gmail Reply-To/action destination, Return-Path mismatch, SPF/DKIM/DMARC anomalies, urgency-based social engineering, and an unrelated tracking resource.

11. Awareness Guidance

Check the complete sender address, inspect Reply-To and destinations, be suspicious of urgent account warnings, verify through the official service directly, and report suspicious messages through the approved organizational channel.

Do not reply, click suspicious links, enter credentials or OTPs after following email links, or trust logos and branding as proof of authenticity.

12. Recommended SOC Controls

Recommended controls include DMARC enforcement, SPF/DKIM monitoring, secure email gateways, sender/Reply-To mismatch detection, lookalike-domain detection, URL reputation scanning, phishing-awareness training, easy reporting mechanisms, SIEM correlation, and quarantine/blocking of confirmed malicious infrastructure.

13. Limitations

This is static analysis. The report does not claim that extracted domains are currently active, that the sender IP is definitively attacker-controlled, that the claimed login IP belongs to an attacker, or that a victim clicked the

message. Those conclusions require additional controlled threat-intelligence or incident-response evidence.

14. Conclusion

Sample-01 demonstrates a realistic phishing pattern combining trusted-brand impersonation, urgency, mismatched sender identities, authentication anomalies, and an unrelated action destination. The message should be treated as HIGH-RISK PHISHING with HIGH confidence.

15. Key Indicator Table

Indicator	Evidence	Severity
Brand impersonation	Microsoft account team	High
From mismatch	access-accsecurity.com	Critical
Reply-To mismatch	sotrecognizd@gmail.com	Critical
Return-Path mismatch	thcultarfdes.co.uk	High
Authentication	SPF none; DKIM none; DMARC permerror	High
Urgency/fear	Unusual sign-in; Russia/Moscow	High
Action destination	Report The User → Gmail	Critical
External resource	thebandalisty[.]com	Medium

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